

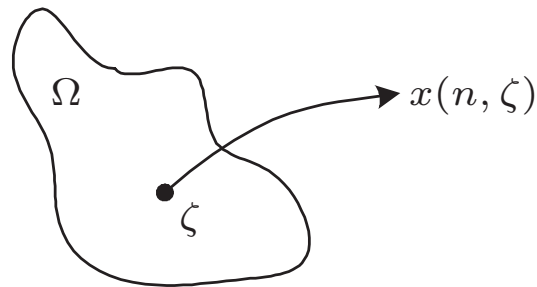
Stochastic Signals Overview

- Definitions
- Second order statistics
- Stationarity and ergodicity
- Random signal variability
- Power spectral density
- Linear systems with stationary inputs
- Random signal memory
- Correlation matrices

Introduction

- Discrete-time **stochastic processes** provides a mathematical framework for working with non-deterministic signals
- Signals that have an exact functional relationship are often called **predictable** or **deterministic**, though some stochastic processes are predictable
- I'm going to use the term **deterministic** to refer to signals that are not affected by the outcome of a random experiment
- I will use the terms **stochastic process** and **random process** interchangeably

Probability Space



- Conceptually we should imagine a sample space with some number (possibly infinite) of outcomes: $\Omega = \{\zeta_1, \zeta_2, \dots\}$
- Each has a probability $\Pr\{\zeta_k\}$
- By some rule, each outcome generates a sequence $x(n, \zeta_k)$
- We can think of $x(n, \zeta_k)$ as a vector of (possibly) infinite duration
- Note that the *entire sequence* is generated from a *single outcome* of the underlying experiment
- $x(n, \zeta)$ is called a **discrete-time stochastic process** or a **random sequence**

Definitions and Interpretations

- Interpretations
 - **Random variable:** $x(n, \zeta)$ with $n = n_o$ fixed and ζ treated as a variable
 - **Sample Sequence:** $x(n, \zeta)$ with $\zeta = \zeta_k$ fixed and n treated as an independent (non-random) variable
 - **Number:** $x(n, \zeta)$ with both $\zeta = \zeta_k$ and $n = n_o$ fixed
 - **Stochastic Process:** $x(n, \zeta)$ with both ζ and n treated as variables
- **Realization:** a sample sequence
- **Ensemble:** The set of all possible sequences, $\{x(n, \zeta)\}$

Probability Functions

In order to fully characterize a stochastic process, we must consider the CDF or PDF

$$F_x(x_1, \dots, x_k; n_1, \dots, n_k) = \Pr \{x(n_1) \leq x_1, \dots, x(n_k) \leq x_k\}$$
$$f_x(x_1, \dots, x_k; n_1, \dots, n_k) = \frac{\partial^k F_x(x_1, \dots, x_k; n_1, \dots, n_k)}{\partial x_1 \dots \partial x_k}$$

for every $k \geq 1$ and any set of sample times $\{n_1, n_2, \dots, n_k\}$.

- Without additional sweeping assumptions, estimation of $f_x(\cdot)$ from a realization is impossible
- Many stochastic processes can be characterized accurately or, at least, usefully by much less information
- To simplify notation, from here on will mostly use $x(n)$ to denote both random processes and single realizations
- In most cases will assume $x(n)$ is complex valued

Second Order Statistics

At any time n , we can specify the mean and variance of $x(n)$

$$\mu_x(n) \triangleq \text{E}[x(n)] \qquad \sigma_x^2(n) \triangleq \text{E}[|x(n) - \mu_x(n)|^2]$$

- $\mu_x(n)$ and $\sigma_x^2(n)$ are both deterministic sequences
- The expectation is taken over the ensemble.
- In general, the second-order statistics at two different times are given by the autocorrelation or autocovariance sequences.

- **Autocorrelation Sequence**

$$r_{xx}(n_1, n_2) = \text{E}[x(n_1)x^*(n_2)]$$

- **Autocovariance Sequence**

$$\begin{aligned} \gamma_{xx}(n_1, n_2) &= \text{E}[(x(n_1) - \mu_x(n_1))(x(n_2) - \mu_x(n_2))^*] \\ &= r_{xx}(n_1, n_2) - \mu_x(n_1)\mu_x^*(n_2) \end{aligned}$$

Cross-Correlation and Cross-Covariance

Cross-Correlation

$$r_{xy}(n_1, n_2) = \text{E}[x(n_1)y^*(n_2)]$$

Cross-Covariance

$$\begin{aligned}\gamma_{xy}(n_1, n_2) &= \text{E}[(x(n_1) - \mu_x(n_1))(y(n_2) - \mu_y(n_2))^*] \\ &= r_{xy}(n_1, n_2) - \mu_x(n_1)\mu_y^*(n_2)\end{aligned}$$

Normalized Cross-Correlation

$$\rho_{xy}(n_1, n_2) = \frac{\gamma_{xy}(n_1, n_2)}{\sigma_x(n_1)\sigma_y(n_2)}$$

More Definitions

- **Independent:** iff

$$f_x(x_1, \dots, x_k; n_1, \dots, n_k) = \prod_{\ell=1}^k f_{\ell}(x_{\ell}; n_{\ell}) \quad \forall k$$

- **Uncorrelated:** if

$$\gamma_x(n_1, n_2) = \begin{cases} \sigma_x^2(n_1) & n_1 = n_2 \\ 0 & n_1 \neq n_2 \end{cases}$$

- **Orthogonal:** if

$$r_x(n_1, n_2) = \begin{cases} \sigma_x^2(n_1) + |\mu_x(n_1)|^2 & n_1 = n_2 \\ 0 & n_1 \neq n_2 \end{cases}$$

Still More Definitions

- **Wide-sense Periodic:** if

$$\begin{aligned}\mu_x(n) &= \mu_x(n + N) \quad \forall n \\ r_x(n_1, n_2) &= r_x(n_1 + N, n_2) = r_x(n_1, n_2 + N) \\ &= r_x(n_1 + N, n_2 + N)\end{aligned}$$

- **Statistically Independent:** iff for every n_1 and n_2

$$f_{xy}(x, y; n_1, n_2) = f_x(x; n_1) f_y(y; n_2)$$

- **Uncorrelated:** if for every n_1 and n_2 ,

$$\gamma_{xy}(n_1, n_2) = 0$$

- **Orthogonal:** if for every n_1 and n_2

$$r_{xy}(n_1, n_2) = 0$$

Stationarity

Stationarity of Order N : A stochastic process $x(n)$ such that

$$f_x(x_1, \dots, x_N; n_1, \dots, n_N) = f_x(x_1, \dots, x_N; n_1 + k, \dots, n_N + k)$$

for any value for any k .

- Any stochastic process of Order N , is also a stochastic process of order M for all $M \leq N$
- **Strict-Sense Stationary (SSS):** A stochastic process that is stationary of all orders N

Wide Sense Stationary

$$f_x(x_1, x_2; n_1, n_2) = f_x(x_1, x_2; n_1 + k, n_2 + k)$$

- **Wide-Sense Stationary (WSS):** A stochastic process with a constant mean and autocorrelation that only depends on the delay between the two sample times
- WSS Properties

$$E[x(n)] = \mu_x$$

$$r_x(n_1, n_2) = r_x(\ell) = r_x(n_1 - n_2) = E[x(n + \ell)x^*(n)]$$

$$\gamma_x(\ell) = r_x(\ell) - |\mu_x|^2$$

- This implies the variance is also constant, $\text{var}[x(n)] = \sigma_x^2$
- All processes that are stationary of order 2 are WSS
- Not all WSS processes are stationary of order 2
- Note this is slightly different from the text

Example 1: Stationarity

Describe a random process that is stationary. Describe a second random process that is not stationary.

Stationarity Notes

- SSS implies WSS
- If the marginal PDF of a signal is Gaussian for all n , then WSS implies SSS
- The book states that most WSS processes are SSS. True?
- **Jointly Wide-Sense Stationary:** two random signals $x(n)$ and $y(n)$ are jointly WSS if they are both WSS and

$$r_{xy}(\ell) = r_{xy}(n_1 - n_2) = r_{xy}(\ell) = \mathbb{E}[x(n)y^*(n - \ell)]$$

$$\gamma_{xy}(\ell) = \gamma_{xy}(n_1 - n_2) = \gamma_{xy}(\ell) = r_{xy}(\ell) - \mu_x \mu_y^*$$

- WSS is a very useful property because it enables us to consider a spectral description
- In practice, we only need the signal to be WSS long enough to estimate the autocorrelation or cross-correlation

Autocorrelation Sequence Properties

$$r_x(0) = \sigma_x^2 + |\mu_x|^2$$

$$r_x(0) \geq |r_x(\ell)|$$

$$r_x(\ell) = r_x^*(-\ell)$$

$$\sum_{k=1}^M \sum_{m=1}^M \alpha_k r_x(k-m) \alpha_m^* \geq 0 \quad \forall \alpha \text{ sequences}$$

- **Average DC Power:** $|\mu_x|^2$
- **Average AC Power:** σ_x^2
- **Nonnegative Definite:** A sequence is said to be nonnegative definite if it satisfies this last property
- **Positive Definite:** Any sequence that satisfies the last inequality strictly for any α

Comments on Stationarity

- Many real processes are nonstationary
- Best case: can determine from domain knowledge of the process
- Else: must rely on statistical methods
- Many nonstationary processes are approximately locally stationary (stationary over short periods of time)
- Much of *time-frequency* analysis is dedicated to this type of signal
- There is no general mathematical framework for analyzing nonstationary signals
- However, many nonstationary stochastic processes can be understood through linear estimation (i.e., Kalman filters)
- Note that nonstationary is a negative definition: not stationary

Introduction to Ergodicity

- In most practical situations we can only observe one or a few realizations
- If the process is **ergodic**, we can know all statistical information from a single realization
- **Ensemble Averages:** Repeat the experiment many times
- **Time Averages:**

$$\langle f[x(n)] \rangle \triangleq \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N f[x(n)]$$

where $f[x(n)]$ is any function of a stochastic process

Time Averages of Interest

$$\text{Mean value} = \langle x(n) \rangle$$

$$\text{Mean square} = \langle |x(n)|^2 \rangle$$

$$\text{Variance} = \langle |x(n) - \langle x(n) \rangle|^2 \rangle$$

$$\text{Autocorrelation} = \langle x(n)x^*(n - \ell) \rangle$$

$$\text{Autocovariance} = \langle [x(n) - \langle x(n) \rangle][x(n - \ell) - \langle x(n) \rangle]^* \rangle$$

$$\text{Cross-correlation} = \langle x(n)y^*(n - \ell) \rangle$$

$$\text{Cross-covariance} = \langle [x(n) - \langle x(n) \rangle][y(n - \ell) - \langle y(n) \rangle]^* \rangle$$

- Similar to correlation sequences for deterministic power signals
- Both quantities have the same properties
- Difference
 - Time averages are random variables (functions of the experiment outcome)
 - In the deterministic case the quantities are fixed numbers

Ergodic Random Processes

- **Ergodic Random Process:** a random signal for which the ensemble averages *equal* the corresponding time averages
- Like stationarity, there are various degrees
- **Ergodic in the Mean:** a random process such that

$$\langle x(n) \rangle = \mathbb{E}[x(n)] = \mu_x$$

- **Ergodic in Correlation:** a random process such that

$$\langle x(n)x^*(n - \ell) \rangle = \mathbb{E}[x(n)x^*(n - \ell)] = r_x(\ell)$$

- If a process is ergodic in both mean and correlation, it is also WSS
- Only stationary signals can be ergodic
- WSS does not imply any type of ergodicity
- Text: “Almost all stationary processes are also ergodic” True?
- Our usage: **ergodic** = ergodic in both the mean and correlation

More on Ergodicity

- **Joint Ergodicity:** two random signals are jointly ergodic iff they are individually ergodic and

$$\langle x(n)y^*(n - \ell) \rangle = \mathbb{E}[x(n)y^*(n - \ell)]$$

- Stationarity ensures time invariance of the statistics
- Ergodicity implies the statistics can be obtained from a single realization with time averaging
 - One realization (a single ζ_k) is sufficient to estimate any statistic of the underlying random process
- Our standing assumption will be that the random process of interest is WSS and ergodic
- However, you should be able to discuss this within the context of study design

Problems with Ergodicity

- Problem we never know $x(n)$ for $n = -\infty$ to $+\infty$
- In all real situations, we only have finite records
- The most common estimator is then

$$\langle f[x(n)] \rangle_N \approx \frac{1}{2N+1} \sum_{n=-N}^N f[x(n)]$$

- Note that it is a random variable
- How good is it?
 - Bias
 - Variance
 - Consistent
 - Confidence intervals
 - Distribution
- This is one of the key topics of this class

Ergodic Processes vs. Deterministic Signals

$$r_x(\ell) = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)x^*(n-\ell)$$

- The autocorrelation of a deterministic power signal and a ergodic process can be calculated with the same infinite summation
- What's the difference then?
 - With deterministic signals there is only one signal
 - With stochastic signals, we assume it was generated from an underlying random experiment ζ_k
 - This enables us to consider the ensemble of possible signals:
 $r_x(\ell) = E[x(n)x^*(n-\ell)]$
 - We can therefore draw inferences and make predictions about the population of possible outcomes, not merely this one signal
- Whether you define a given signal as deterministic or as a single realization of a random process depends largely on the application

Random Processes in the Frequency Domain

Power Spectral Density (PSD)

$$R_x(e^{j\omega}) \triangleq \mathcal{F}\{r_x(\ell)\} = \sum_{\ell=-\infty}^{\infty} r_x(\ell)e^{-j\omega\ell}$$

$$r_x(\ell) = \mathcal{F}^{-1}\{R_x(e^{j\omega})\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} R_x(e^{j\omega})e^{j\omega\ell} d\omega$$

- Stationary random processes have deterministic correlation sequences
- They have a single index (independent variable)
- Note again that the power spectral density can be calculated with the same equation for deterministic and ergodic signals

Periodic and Non-Periodic Processes

$$R_x(e^{j\omega}) \triangleq \mathcal{F}\{r_x(\ell)\} = \sum_{\ell=-\infty}^{\infty} r_x(\ell)e^{-j\omega\ell}$$

- If $r_x(\ell)$ is periodic, the DTFS is most appropriate
- **Line Spectrum** If we allow impulses in the PSD, then the PSD of a periodic $r_x(\ell)$ consists of an impulse train
- If the process $x(n)$ is non-zero mean (i.e., nonzero average DC power), the PSD will contain an impulse at $\omega = 0$
- More generally, a random process can be composed of both deterministic components and non-periodic components

Power Spectral Density Properties

- $R_x(e^{j\omega})$ is real-valued
- $R_x(e^{j\omega})$ is periodic with period 2π
- $R_x(e^{j\omega}) \geq 0$ is nonnegative
- $R_x(e^{j\omega})$ has nonnegative area with average value given by

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} R_x(e^{j\omega}) d\omega = r_x(0) = \mathbb{E}[|x(n)|^2]$$

- If $x(n)$ is real-valued
 - $r_x(\ell)$ is real and even
 - $R_x(e^{j\omega})$ is an even function of ω
- What if $x(n)$ is complex-valued?

White Noise

White Noise Process: A random WSS sequence $w(n)$ such that

$$\mathbb{E}[w(n)] = \mu_w \qquad r_w(\ell) = (\sigma_w^2 + \mu_w^2) \delta(\ell)$$

- Specifically, this is a *second-order* white process
 - Notation: $w(n) \sim \text{WN}(\mu_w, \sigma_w^2)$
- Not a complete characterization of $w(n)$: the marginal PDF could be anything
- If $w(n)$ is Gaussian, then a white Gaussian process is denoted by $w(n) \sim \text{WGN}(\mu_w, \sigma_w^2)$
- The term *white* comes from properties of white light

Harmonic Process

Harmonic Process any process defined by

$$x(n) = \sum_{k=1}^M a_k \cos(\omega_k n + \phi_k) \quad \omega_k \neq 0$$

where M , $\{a_k\}_1^M$, and $\{\omega_k\}_1^M$ are constant. The random variables $\{\phi_k\}_1^M$ are pairwise independent and uniformly distributed in the interval $[-\pi, \pi]$.

- $x(n)$ is stationary and ergodic with zero mean and autocorrelation

$$r_x(\ell) = \frac{1}{2} \sum_{k=1}^M a_k^2 \cos(\omega_k \ell)$$

- Note the cosines in the autocorrelation are in-phase

Harmonic Process PSD

$$x(n) = \sum_{k=1}^M a_k \cos(\omega_k n + \phi_k) \quad \omega_k \neq 0$$

- The PSD consists of pairs of impulses (line spectrum) of area $\frac{\pi a_k^2}{2}$ located at frequencies $\pm\omega_k$

$$R_x(e^{j\omega}) = \frac{\pi}{2} \sum_{k=1}^M a_k^2 [\delta(\omega - \omega_k) + \delta(\omega + \omega_k)] \quad -\pi \leq \omega \leq \pi$$

- If all $\omega_k/(2\pi)$ are rational numbers, $x(n)$ is periodic and the impulses are equally spaced apart
- This never happens, unless there is a single periodic (perhaps non-sinusoidal) component
- Otherwise they are *almost periodic* (always happens)

Harmonic Process Comments

$$x(n) = \sum_{k=1}^M a_k \cos(\omega_k n + \phi_k) \quad \omega_k \neq 0$$

- It is only stationary if all of the random phases are equally likely (uniformly distributed over all possible angles)
- This is an unusual circumstance where the signal is stationary but is parameterized by one or more random variables that are constant over all n
- In general, is non-Gaussian
- Is a predictable random sequence! (also highly unusual)

Cross-Power Spectral Density

Cross-power Spectral Density: if $x(n)$ and $y(n)$ are jointly stationary stochastic processes,

$$R_{xy}(e^{j\omega}) \triangleq \mathcal{F}\{r_{xy}(\ell)\} = \sum_{\ell=-\infty}^{\infty} r_{xy}(\ell)e^{-j\omega\ell}$$

$$r_{xy}(\ell) = \frac{1}{2\pi} \int_{-\pi}^{\pi} R_{xy}(e^{j\omega})e^{j\omega\ell} d\omega$$

$$R_{xy}(e^{j\omega}) = R_{yx}^*(e^{j\omega})$$

- Also known as the **cross-spectrum**
- Note that unlike the PSD, it is not real-valued, in general

Coherence

Normalized Cross-Spectrum

$$\mathcal{G}_{xy}(e^{j\omega}) \triangleq \frac{R_{xy}(e^{j\omega})}{\sqrt{R_x(e^{j\omega})R_y(e^{j\omega})}}$$

Also known as the **coherency spectrum** or simply **coherency**. Similar to the correlation coefficient in frequency.

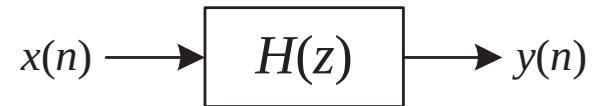
Coherence Function

$$\mathcal{G}_{xy}^2(e^{j\omega}) \triangleq \frac{|R_{xy}(e^{j\omega})|^2}{R_x(e^{j\omega})R_y(e^{j\omega})}$$

Also known as the **coherence** and **magnitude square coherence**.

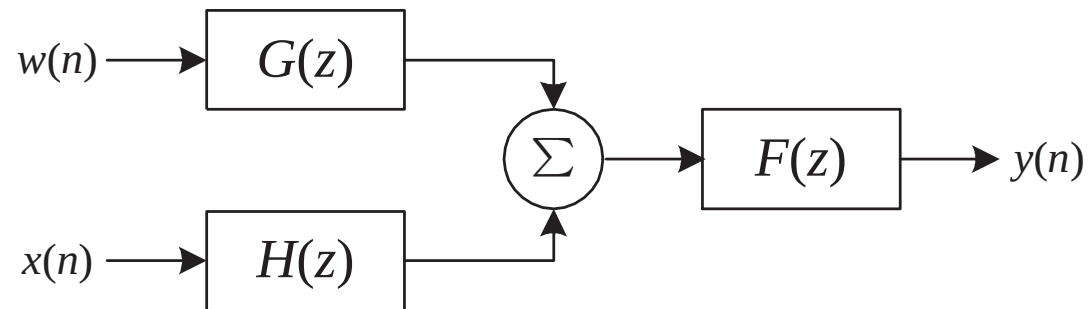
- If $y(n) = h(n) * x(n)$, then $\mathcal{G}_{xy}^2(e^{j\omega}) = 1 \quad \forall \omega$
- If $r_{xy}(\ell) = 0$, then $\mathcal{G}_{xy}^2(e^{j\omega}) = 0 \quad \forall \omega$
- $0 \leq \mathcal{G}_{xy}^2 \leq 1$

Linear Transforms and Coherence



- Linear transforms have no effect on coherence
- Similar to the case of random variables: $y = mx + b$
 - x and y are perfectly correlated: $\rho = \pm 1$

Linear Transforms and Coherence



$$\mathcal{G}_{xy}^2(e^{j\omega}) = \frac{R_x(e^{j\omega})|H(e^{j\omega})|^2}{R_x(e^{j\omega})|H(e^{j\omega})|^2 + R_w(e^{j\omega})|G(e^{j\omega})|^2}$$

- Noise $w(n)$ decreases coherence
- The final linear transform $F(z)$ has no effect!

Complex Spectral Density Functions

Complex Spectral Density

$$R_x(z) = \sum_{k=-\infty}^{\infty} r_x(\ell) z^{-\ell}$$

$$R_y(z) = \sum_{k=-\infty}^{\infty} r_y(\ell) z^{-\ell}$$

Complex Cross-Spectral Density

$$R_{xy}(z) = \sum_{k=-\infty}^{\infty} r_{xy}(\ell) z^{-\ell}$$

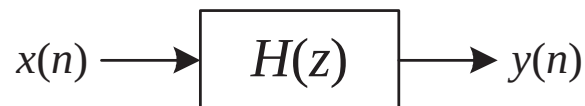
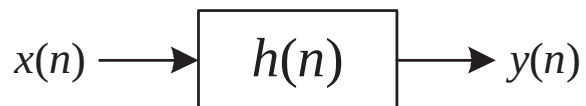
Random Processes and Linear Systems

If the input to an LTI system is a random process, so is the output.

$$y(n, \zeta) = \sum_{k=-\infty}^{\infty} h(k)x(n-k, \zeta)$$

- If the system is BIBO stable and the input process is stationary with $E[|x(n, \zeta)|] < \infty$, then the output converges absolutely with probability one.
- In English, the output is stationary
- If $E[|x(n, \zeta)|^2] < \infty$, then $E[|y(n, \zeta)|^2] < \infty$
- If the $h(n)$ has finite energy, the output converges in the mean square sense

Linear System Statistics



Let $x(n)$ be a random process that is the input to an LTI system with an output $y(n)$.

$$\mu_y = \sum_{k=-\infty}^{\infty} h(k) \mathbb{E}[x(n-k)] = \mu_x \sum_{k=-\infty}^{\infty} h(k) = \mu_x H(e^{j0})$$

$$r_{xy}(\ell) = \sum_{k=-\infty}^{\infty} h^*(k) r_x(\ell+k) = \sum_{m=-\infty}^{\infty} h^*(-m) r_x(\ell-m)$$

$$r_{xy}(\ell) = h^*(-\ell) * r_x(\ell)$$

$$r_{yx}(\ell) = h(\ell) * r_x(\ell)$$

$$r_y(\ell) = h(\ell) * r_{xy}(\ell) = h(\ell) * h^*(-\ell) * r_x(\ell) = r_h(\ell) * r_x(\ell)$$

Output Power



Let $x(n)$ be a random process that is the input to an LTI system with an output $y(n)$.

$$\begin{aligned} P_y &= r_y(0) = [r_h(\ell) * r_x(\ell)]_{\ell=0} \\ &= \sum_{k=-\infty}^{\infty} r_h(k) r_x(-k) \\ &= \sum_{k=-\infty}^{\infty} r_h(k) r_x^*(k) \end{aligned}$$

If the system is FIR, then

$$P_y = \mathbf{h}^H \mathbf{R}_x \mathbf{h}$$

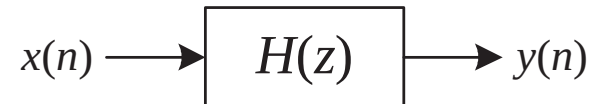
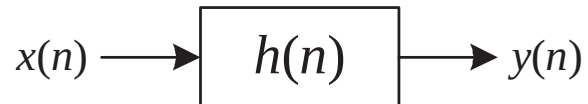
If $\mu_x = 0$, then $\mu_y = 0$ and $\sigma_y^2 = P_y$

Output Distribution



- In general, it is very difficult to solve for the output PDF (even when $y(n)$ is WSS)
- If $x(n)$ is a Gaussian process, the output is a Gaussian process
- If $x(n)$ is IID,
 - The output is a weighted sum of IID random variables
 - If the distribution of $x(n)$ is stable, then $y(n)$ has the same distribution (even if the mean and variance differ)
 - If many of the largest weights are approximately equal so that many elements of the input signal have an equal effect on the output, then the CLT applies (approximately) and the output will be approximately Gaussian

z Domain Analysis



$$\mathcal{Z} \{h^*(-n)\} = H^*(z^{-*})$$

$$R_{xy}(z) = \mathcal{Z} \{h^*(-\ell) * r_x(\ell)\} = H^*(z^{-*})R_x(z)$$

$$R_{yx}(z) = \mathcal{Z} \{h(\ell) * r_x(\ell)\} = H(z)R_x(z)$$

$$R_y(z) = H(z)H^*(z^{-*})R_x(z)$$

Note that if $h(n)$ is real, then

$$h^*(-n) = h(-n)$$

$$h(-n) \xleftrightarrow{\mathcal{Z}} H(z^{-1})$$

Frequency Domain Analysis



If the system is stable, $z = e^{j\omega}$ lies in the ROC and the following relations hold

$$R_{xy}(e^{j\omega}) = H^*(e^{j\omega})R_x(e^{j\omega})$$

$$R_{yx}(e^{j\omega}) = H(e^{j\omega})R_x(e^{j\omega})$$

$$R_y(e^{j\omega}) = |H(e^{j\omega})|^2 R_x(e^{j\omega})$$

- Knowing $r_y(\ell)$ and $r_x(\ell)$ or the input and output PSD's are sufficient to determine $|H(e^{j\omega})|$
- We can't estimate $\angle H(e^{j\omega})$ from this information (the second order statistics)
- Only $r_{xy}(\ell)$ or $R_{xy}(e^{j\omega})$ can provide phase information

Random Signal & System Memory



- **Zero-memory:** a process for which $r_x(\ell) = \sigma_x^2 \delta(\ell)$
- Examples: white noise, IID process
- We can create a signal with memory (dependence) by passing a zero-memory process through an LTI system
- Extent and degree of imposed dependence depend on $h(n)$

Correlation Length

- **Correlation Length:** given a WSS process,

$$L_c = \frac{1}{r_x(0)} \sum_{\ell=0}^{\infty} r_x(\ell) = \sum_{\ell=0}^{\infty} \rho_x(\ell)$$

- Equal to the “area” of the normalized autocorrelation curve
- Undesirable properties
 - Why is it one sided?
 - Lengths should not be negative, in general. Could this be negative?

$$r(\ell) = [1.0000, -0.3214, -0.7538] \text{ for } \ell = 1, 2, 3$$

- “zero-memory” processes have a non-zero correlation length ($L_c = 1$)

Short Memory Processes

- **Short Memory:** a WSS process $x(n)$ such that

$$\sum_{\ell=-\infty}^{\infty} \rho_x(\ell) < \infty$$

- For example, autocorrelation decays exponentially

$$\rho_x(\ell) \approx a^{|\ell|} \text{ for large } \ell$$

Long Memory Processes

- **Long memory:** for WSS signal $x(n)$ with finite variance, if there exists $0 < \alpha < 1$ and $C_r > 0$

$$\lim_{\ell \rightarrow \infty} \frac{1}{C_r \sigma_x^2} r_x(\ell) \ell^{-\alpha} = 1$$

- Equivalently, there exists $0 \leq \beta < 1$ and $C_r > 0$ such that

$$\lim_{\omega \rightarrow 0} \frac{1}{C_r \sigma_x^2} R_x(e^{j\omega}) |\omega|^\beta = 1$$

- Implies
 - The autocorrelation has heavy tails
 - Autocorrelation decays as a power law

$$\sum_{\ell=-\infty}^{\infty} \rho_x(\ell) = \infty \quad \rho_x(\ell) \approx C_r |\ell|^{-\alpha} \text{ as } \ell \rightarrow \infty$$

- Has infinite autocorrelation length

Correlation Matrices

Let the random vector $\mathbf{x}(n)$ be related to the (possibly nonstationary) random process $x(n)$ as follows

$$\begin{aligned}\mathbf{x}(n) &\triangleq [x(n) \quad x(n-1) \quad \cdots \quad x(n-M+1)]^H \\ \mathbf{E}[\mathbf{x}(n)] &= [\mu_x(n) \quad \mu_x(n-1) \quad \cdots \quad \mu_x(n-M+1)]^H \\ \mathbf{R}_x(n) &= \mathbf{E}[\mathbf{x}(n)\mathbf{x}(n)^H] \\ &= \begin{bmatrix} r_x(n, n) & \cdots & r_x(n, n-M+1) \\ \vdots & \ddots & \vdots \\ r_x(n-M+1, n) & \cdots & r_x(n-M+1, n-M+1) \end{bmatrix}\end{aligned}$$

Note that $\mathbf{R}_x(n)$ is nonnegative definite and Hermitian since $r_x(n-i, n-j) = r_x^*(n-j, n-i)$.

Correlation Matrices

If $x(n)$ is a stationary process, the correlation matrix becomes

$$\mathbf{R}_x(n) = \begin{bmatrix} r_x(0) & r_x(1) & r_x(2) & \cdots & r_x(M-1) \\ r_x^*(1) & r_x(0) & r_x(1) & \cdots & r_x(M-2) \\ r_x^*(2) & r_x^*(1) & r_x(0) & \cdots & r_x(M-3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_x^*(M-1) & r_x^*(M-2) & r_x^*(M-3) & \cdots & r_x(0) \end{bmatrix}$$

In this case $\mathbf{R}_x$ is Hermitian, $\mathbf{R}_x = \mathbf{R}_x^H$, Toeplitz (the elements along each diagonal are equal), and nonnegative definite.

Conditioning of Correlation Matrix

- **Condition number:** of a positive definite matrix $\mathbf{R}_x$ is

$$\chi(\mathbf{R}_x) \triangleq \frac{\lambda_{\max}}{\lambda_{\min}}$$

where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and smallest eigenvalues of the autocorrelation matrix, respectively

- If $x(n)$ is a WSS random process, then the eigenvalues of the autocorrelation matrix are bounded by the dynamic range of the PSD

$$\min_{\omega} R(e^{j\omega}) \leq \lambda_i \leq \max_{\omega} R(e^{j\omega}) \quad \forall \lambda_i$$

- See text for proof
- Interpretation: a large spread in eigenvalues implies
 - PSD is more variable (less flat)
 - Process is less like white noise (more predictable)